ResQ-MARL: DISASTER RESCUE BLUEPRINT

Hierarchical AI Behaviors, Earthquake Obstacles, Aesthetics & Team Action Plan

Target Scenario: Post-Earthquake Urban Search & Rescue (USAR) | **Framework:** Unity C# + ML-Agents | **Status:** Complete Blueprint

Executive Blueprint Summary:
*This master document provides the practical implementation guide to transform Arena into an **Autonomous Disaster Response & USAR Testbed**. It introduces **Hierarchical AI** (ML-Agents locomotion + C# rescue state machine) so survivors follow drones to safety without retraining models, specifies **4 realistic earthquake debris types** (Fallen Pillars, Tilted Slabs, Rubble Mounds, Barricades), delivers an emergency visual makeover, and partitions the work into clear responsibilities for a 3-person team.*

1. Direct Codebase Mapping: Entertainment Game to Disaster System

Codebase Script	Previous Game Role	New Disaster Rescue Role	Key Code Action
HideAndSeekAgent.cs	Seekers hunting hiders	Autonomous Rescue Swarm	Drones sweep separate sectors to avoid dogpiling.
AgentActions.cs	Hiders freeze on tag	Trapped Survivors	Add states: Trapped -> Escorting -> Rescued.
GameController.cs	Check tag count	Mission Controller	Trigger win when all victims reach Safe Triage Zone.
BoxHolding.cs	Grab/lock boxes	Debris Manipulator	Robots push/clear fallen beams to open paths.
ArchitectManager.cs	Spawn walls & boxes	Site Damage Mapper	Add tools for Pillars, Tilted Slabs, and Rubble.
ArchitectUI.cs	Build HUD	Incident Command HUD	Show Victims Located and Evacuated counters.

2. Hierarchical AI: Changing AI Behavior Without Retraining

• **Low-Level Locomotion Layer:** Existing trained ONNX neural networks handle continuous physics movement, room navigation, and obstacle avoidance.

• **High-Level Mission State Machine (C#):** Overrides agent behavior based on mission events:

- **State 1: Trapped / Sheltered:** Survivors shelter under debris using trained hiding weights.

- **State 2: Being Escorted:** When a rescue drone arrives within 1.2m, the survivor stops running away and follows the drone toward the Safe Triage Zone.

- **State 3: Safely Evacuated:** Upon touching the green Safe Zone collider, the survivor freezes, turns green, and registers in the mission triage counter.

- **Swarm Sector Retasking:** When Drone A claims Victim 1, Drones B & C divert to unexplored rooms, eliminating redundant chasing.

3. Four Realistic Earthquake Debris Obstacles

Obstacle Type	Unity Construction	Dimensions / Scale	Disaster Rescue Role
1. Collapsed Pillar	Stretched Cylinder or heavy Cube	(0.8, 0.8, 5.0) laying horizontally	Heavy fallen structural column blocking doorway.
2. Tilted Slab / Ramp	Use existing Ramp.prefab	(2.5, 1.2, 4.0) angled at 25-35°	Broken ceiling slab forming crawlspace for victims.
3. Rubble Mound	Compound of 3-4 rotated concrete cubes	(1.5, 0.7, 1.8) irregular pile	Fractured concrete heap blocking rover vision.
4. Hazard Barricade	Thin wall with yellow/black hazard tape	(3.0, 1.2, 0.2) partition	Emergency barrier placed to seal dangerous zones.

4. Three-Person Role Division & Work Plan

Member	Core Focus	Exact Implementation Tasks
Person 1 **AI & Rescue Lead**	**Triage Zone & Escort AI**	• Add SafeTriageZone.cs trigger to arena floor. • Attach SurvivorRescueState.cs to hiders to enable follow/escort logic. • Update GameController.cs win condition to all victims evacuated.
Person 2 **3D Debris Lead**	**Earthquake Debris & Arena**	• Build Pillar, Slab/Ramp, Rubble Mound, and Barricade prefabs. • Add new debris tools (keys 1-5) into ArchitectManager.cs. • Build 1 ready-to-demo Collapsed Building layout preset.
Person 3 **Visual & HUD Lead**	**Visual Reskin & Mission HUD**	• Reskin drones (Hazard Orange + flashing beacon) and survivors (distress pulse). • Redesign ArchitectUI.cs into Incident Command HUD with live triage counts. • Retexture arena floor to cracked concrete with green Safe Base perimeter.

5. Step-by-Step Implementation Timeline

Day 1 (AI & Safe Zone): Add SafeTriageZone.cs and SurvivorRescueState.cs. Test that survivors follow drones to safety upon touch.

Day 2 (Debris Prefabs): Create the 4 earthquake debris prefabs and hook them into ArchitectManager.cs tools 1-5.

Day 3 (Aesthetics & HUD): Apply orange drone materials, searchlight scanner, survivor badges, and emergency HUD counters.

Day 4 (Integration & Rehearsal): Load the collapsed building preset and rehearse the 5-minute defense demo.